Disease detectives: follow the cause

Teacher guide and worked answers

Distinguish communicable and non-communicable disease using cause and transmission evidence.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Spring | C33

Describe communicable & non-communicable disease.

Directly develops the Week 6 outcome using Pearson 5.2 and 5.4-5.6. This exemplar selects representative infections; it does not replace all Topic 5 disease examples.

Prior learning

A microorganism is microscopic; not all microorganisms cause disease.

A cause differs from an observed symptom.

Success evidence

Classify a disease from its cause and transmission route.

Explain an indirect transmission route and how an intervention could interrupt it.

Explain why a shared symptom or risk factor is not proof of a diagnosis.

Equipment

Printed shared evidence and one selected response route; pen, ruler and calculator.

Projector or offline HTML/PowerPoint. Optional word processor or spreadsheet.

Preparation

Print the shared evidence page, one chosen response route and exit page. Routes are alternatives, not a workload ladder.

Read the teacher answers before teaching. All numerical sample data in this lesson are constructed for learning.

Practical care

Paper and screen tasks; no biological sampling, cultures, medicine handling or personal health disclosures.

Ready to teach alternative

All evidence is included. Point, annotate, dictate or discuss quietly instead of writing or using a device.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	One symptom. Two explanations.

Slide	Stage	Minutes	Focus
2	Retrieval	2-5	Cause, symptom or risk?
3	I do	5-9	I follow the causal evidence
4	I do	9-12	Transmission can be indirect
5	I do	12-15	A careful explanation of risk
6	We do	15-18	Case-board challenge
7	We do	18-20	Interrupt one route
8	We do	20-22	Repair the false headline
9	Hinge	22-25	Which statement is scientifically sound?
10	You do	25-26	Build an evidence-based explanation
11	You do	26-36	Independent case report
12	Review	36-38	Let evidence change a decision
13	Review	38-39	A defensible case report
14	Exit	39-40	Three scientific distinctions
15	Exit	Within stage	Your next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening One symptom. Two explanations.

Ask whether a cough proves an infection. Hold the question open until the two cause cards are compared. We classify supplied examples; we do not classify classmates.

2 Retrieval Cause, symptom or risk?

R1 no; many microorganisms are harmless or useful. R2 no; the same symptom may have different causes. R3 no. If unsure, use an everyday chance example without asking for health stories.

3 I do I follow the causal evidence

Model: “A says bacteria can spread through air, so I classify tuberculosis as communicable. D says asthma is not transmitted. Both can involve coughing, so cough alone cannot decide.”

4 I do Transmission can be indirect

Think aloud: “The protist causes malaria. The mosquito carries it. Therefore I need not draw a direct touch between people. I must not call the mosquito the pathogen.” The simplified route omits the parasite life cycle and does not predict whether an individual bite infects.

5 I do A careful explanation of risk

Model: “The heart-disease card gives several influences. It cannot tell me whether a named person will develop disease. A claim blaming one behaviour ignores genes, environment and access to resources.” No personal disclosures.

6 We do Case-board challenge

Give silent think time. Invite one choice plus reason, then ask another pupil to test the reason with a counterexample. A/B/C/F communicable; D/E non-communicable. Use printable cards or interactive reveal, same science.

7 We do Interrupt one route

Cholera: safer water/food reduces exposure to water/food-borne bacteria. Malaria: mosquito barriers such as insecticide-treated nets can reduce vector bites, depending on use/context. These are mechanism examples, not travel advice. Ask what part of the route is interrupted.

8 We do Repair the false headline

Expected: cough can have different causes; tuberculosis is communicable, asthma is not. Evidence about cause is needed. Ask the audience to say which exact word was too strong. No actual diagnosis is attempted.

Reveal: The same symptom can have different causes. These two cause cards belong in different categories.

9 Hinge Which statement is scientifically sound?

Correct B. If pupils select A, return to vector diagram; if C, compare cause and symptom; if D, revisit risk diagram.

A. Malaria is non-communicable because a vector is involved. - A pathogen can spread indirectly through a vector; malaria is communicable.

B. A communicable disease has an infectious cause that can spread. - Yes. Spread can be direct or indirect, including air, water and vectors.

C. A shared symptom proves the same cause. - Tuberculosis and asthma can both involve coughing but have different causes.

D. One risk factor guarantees disease. - A risk factor affects probability; it does not make disease certain.

10 You do Build an evidence-based explanation

Set one route. Pupils produce a case classification, a transmission explanation and a cautious conclusion. Route selection responds to current support needs.

11 You do Independent case report

Supported S1-S4; standard M1-M4; stretch T1-T4. Circulate after first classification. For wrong malaria category, trace person-vector-person without revealing the category. For blame statements, redirect to multiple factors.

12 Review Let evidence change a decision

A partner, teacher or TA is an audience; quiet written review is valid. Ask a genuine challenge; pupil improves or defends using source evidence. Do not publicly rank health knowledge.

13 Review A defensible case report

Share model: “C is communicable: a mosquito can carry the malaria protist between hosts. The mosquito is the vector, not the pathogen.” Ask pupils to amend one term if needed.

14 Exit Three scientific distinctions

E1 transmissible infectious cause. E2 protist; mosquito. E3 different causes can cause a similar symptom. Use responses to plan barrier/immune lesson.

15 Exit Your next useful step

Shared exit minute. Collect one response per pupil, accepting annotated arrows or dictated words. If cause remains insecure, begin next lesson with A versus D.

Answers and assessment

Retrieval R1-R3

R1 no. R2 a symptom is not its cause; different causes may produce a similar symptom. R3 increased chance is not certainty.

W1-W3

W1 A/B/C/F communicable; D/E non-communicable. W2 safer water/food interrupts the cholera ingestion route; a suitable mosquito barrier reduces malaria-vector bites. W3 cough alone cannot establish cause: A and D provide a counterexample.

Hinge A-D

A false: indirect vector spread is communicable. B correct: a transmissible infectious cause defines the category. C false: shared symptoms do not prove shared cause. D false: a risk factor does not guarantee disease.

S1 / M1

A/B/C/F communicable; D/E non-communicable. C: protist carried by mosquito. D: airway condition not transmitted. Examples of different routes: airborne tuberculosis versus contaminated water/food in cholera.

S2 / M2

An infected person can infect a feeding mosquito; an infected mosquito can pass the protist to another person through a bite. Protist = pathogen; mosquito = vector. A barrier reduces opportunities for infectious bites; it does not treat infection.

S3

Safer water/food reduces exposure to cholera bacteria entering through contaminated water/food. A mosquito barrier does not target this transmission route.

S4 / M3

A and D can both involve coughing but have different causes/categories. Cough alone cannot establish the infectious cause or diagnose either condition.

M4

A risk factor can increase the chance of disease, not guarantee it. Genetic/body factors, environment and behaviour/circumstances can combine. Do not accept blame or certainty as an explanation.

T1

G communicable: fungal spores carry the infectious cause through air. H non-communicable: an inherited condition is passed through genetic information at reproduction, not transmitted as an infection.

T2

No. Dust exposure can cause irritation or trigger airway symptoms. Additional evidence could include appropriate clinical/pathogen testing or exposure histories; pupils do not perform these tests. A temporal association alone cannot establish the cause.

T3

False: antibiotics may be used against some bacterial infections; they do not treat influenza viruses. Classification as communicable alone does not determine a treatment. Clinical decisions are outside this task.

T4

Barriers may not remove every exposure and effectiveness depends on use/context; reducing the chance of transmission does not make it impossible.

Exit E1-E3

E1 a disease whose infectious cause can spread directly or indirectly. E2 malaria protist; mosquito vector. E3 similar symptoms can have different causes.

L1 and assessment

Accept any specific evidence-led revision or justified confirmation. Secure core = correct causal classification, vector/pathogen distinction, and no symptom-to-diagnosis leap. Re-teach with A versus D if a pupil uses cough as the category rule.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 10, We do 7, hinge 3, independent 11, review 3, exit 1. Zero-minute slides share their stage time.

Choose one response route. Accept accurate speech, pointing plus explanation, annotations or writing as evidence.

A quiet review still has an audience: teacher or TA reads the pupil's decision, asks one evidence question, and the pupil revises or confirms it.

Keep discussion about the fictional or supplied case cards. Pupils need not disclose illnesses, family history or medication.

Interactive We do: predict the case category, test the malaria route, then repair the symptom claim. The paper equivalent is W1-W3 with cards and arrows; the visual is a conceptual route model, not a simulation of infection risk.

Access and responsive teaching

Supported

Same core scientific goal, with word bank, short prompts, read-aloud or pointing/dictation. Do not assign a route from diagnosis.

Standard

Use the supplied evidence to explain and justify an independent scientific decision.

Stretch

Optional challenge after secure core: evaluate a limitation and transfer the science. Higher-tier-only material, where present, is labelled explicitly.

Regulation

Preview the sequence. Offer solo or partner work, a pass and quiet response. Allow a brief reset and re-entry at the next numbered task; no compulsory disclosure, performance or speed competition.

Misconceptions to check

All disease spreads by touching someone.

Some infectious causes spread through air, water or vectors; non-communicable diseases are not transmitted.

Check: Ask pupils to trace the malaria card without a direct person-to-person arrow.

Two people coughing must have the same infection.

Different causes can produce a similar symptom; symptoms alone cannot establish a cause.

Check: Compare tuberculosis and asthma cards.

A risk factor makes a disease certain.

Risk factors alter probability and may interact; they do not guarantee an outcome.

Check: Replace “will” with a justified risk statement.

Antibiotics treat every pathogen.

Antibiotics target bacterial processes; they do not treat viruses.

Check: Ask whether the influenza card supplies a bacterial target.

Word	Meaning
communicable	A disease whose infectious cause can spread between organisms, directly or indirectly.
non-communicable	A disease that is not transmitted between organisms.
pathogen	A disease-causing agent, such as a bacterium, virus, fungus or protist.
vector	An organism that carries a pathogen between hosts.
risk factor	Something associated with a higher chance of a disease; it does not guarantee it.
symptom	A change experienced by a person that may be associated with illness.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 8 September 2026.

WHO: Noncommunicable diseases

Checked the combination of genetic, physiological, environmental and behavioural factors; no current disease burden figures used. Checked 8 September 2026.

NHS: Antibiotics

Checked that antibiotics can treat some bacterial infections but do not treat viral infections. No prescribing advice is part of this lesson. Checked 8 September 2026.